

Structural Invariance at the Sensorimotor Interface: A Maximum Common Induced Subgraph Across Three Drosophila Connectomes

Yanjin Li · FlyWire Qualification Challenge · github.com/Yanjin-ai/flywireconnectome

Abstract

We ask whether synaptic connectivity itself is structurally invariant across independently prepared connectomes. Using the BANC metadata's NBLAST-based 1:1 correspondence columns (Bates et al. 2025), we align individual neurons across BANC (♀ brain+cord), FAFB (♀ brain) and MANC (♂ nerve cord) and search for the largest set of neurons whose directed induced subgraph is identical (isomorphic) in all three. The result is an $N = 109$ -neuron sensorimotor backbone with 14 conserved directed edges, enriched $65.7\times$ for descending and $24.8\times$ for ascending neurons, and 92.7% conserved across sexes. All reported numbers are reproduced by `src/run_analysis.py` and stored under `results/`.

Methods. Correspondence: BANC metadata `fafb_match` / `manc_match` (NBLAST). Graphs: directed, edge existence (not weight). Search space: 3,414 matched triplets $\rightarrow$ 2,798 present in all three edge lists $\rightarrow$ 987-node giant consensus component. Algorithm: greedy disagreement removal + exhaustive expansion (single implementation in `mcis_connectome.solver`), reported as the best of a 3000-seed multi-start. Optimality checked against an ILP (PuLP/CBC) on 50 subgraphs: mean gap 1.15% (max 10.53%).

Key Results

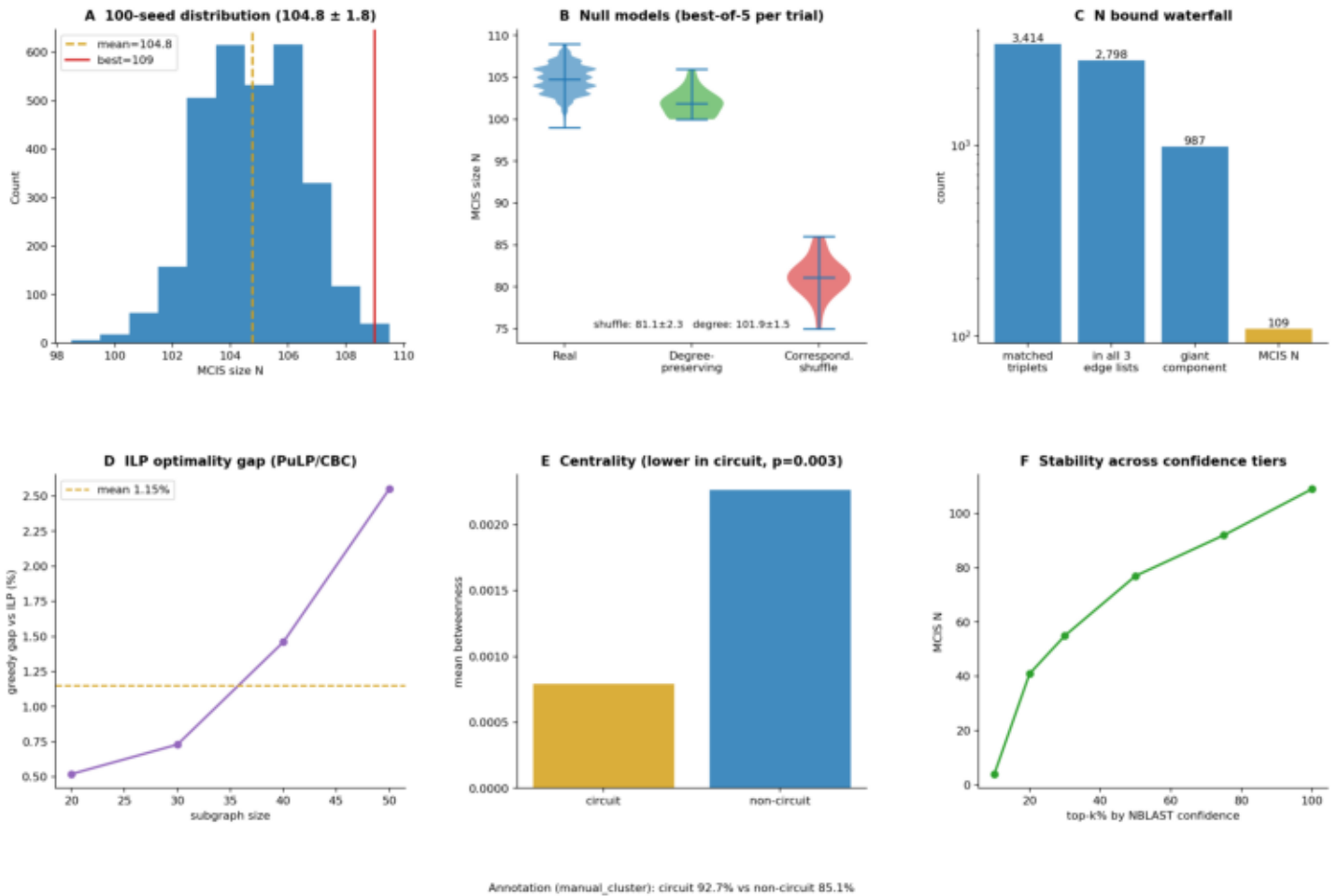
Metric	Value
MCIS size N	109 (dist. 104.8 ± 1.8 , [99,109])
Conserved edges	14
Composition	67 DN + 34 AN + 8 sensory
Descending enrichment	$65.7\times$ ($p=4e-107$)
Ascending enrichment	$24.8\times$ ($p=1e-37$)
Corr-shuffle null	81.1 ± 2.3 ($>15\sigma$ below real)
Degree-preserving null	101.9 ± 1.5 ($\approx$ real mean)
Centrality (lower)	$p = 0.003$ (one-sided)
Annotation quality	92.7% vs 85.1% ($p=0.014$)
Sexual conservation	92.7% (101/109)

Conserved circuit across BANC $\times$ FAFB $\times$ MANC



Robustness, null models, and ILP optimality

MCIS Robustness · Statistical Validation · Optimality



Discussion & Limitations. The conserved backbone is almost entirely descending and ascending neurons (93.3%), giving direct structural support for the sensorimotor bottleneck (Pospisil et al. 2024) and showing it is canalized across sexes and specimens. **PRINCIPAL CAVEAT:** a degree-preserving rewiring null reaches essentially the same MCIS size ($101.9 \pm 1.5 \approx$ real 104.8), so the FAFB degree sequence — not specific edge identity — explains most of the achievable N; NBLAST neuron identity adds the remaining signal (correspondence-shuffle collapses to 81.1). N is a near-optimal heuristic lower bound (ILP gap $\sim 1.15\%$), not a certified global optimum.

Reproduce: `pip install -e . && MCIS_DATA_DIR=... python src/run_analysis.py --seeds 100`